

Gut Microbiome Profiling in Pediatric Ulcerative Colitis

16S rRNA amplicon analysis of PRJNA759642: from raw reads to differential abundance

Davide Maccarrone

1 Executive summary

Objective. To characterize the gut microbiome of children with ulcerative colitis (UC) compared with healthy children, using a reproducible end-to-end 16S rRNA workflow on the public dataset PRJNA759642.

Dataset. 42 fecal samples (19 UC, 23 healthy; age 7–21 years), 16S rRNA V4 amplicons, Illumina MiSeq 2 × 150 bp.

Main findings.

- *Data quality and processing:* after DADA2 processing and chimera removal, the median number of non-chimeric reads per sample was **73,465**, and **73.4%** of all input reads were retained overall.
- *Within-sample diversity:* Shannon diversity was **lower in UC** than in Healthy samples (median Shannon index: 2.44 vs 2.69; Wilcoxon $p = \mathbf{0.002}$). After adjustment for age and gender, Shannon diversity in UC was **0.80-fold** that of Healthy samples (95% CI: **0.71–0.90**). The group difference remained significant at every rarefaction depth tested (1,000–100,000 reads; Wilcoxon $p = \mathbf{0.002}$) and after adjustment for library size.
- *Between-sample diversity:* disease status explained **9.7%** of Bray-Curtis variation (PERMANOVA $p = \mathbf{0.001}$); dispersion did not differ significantly between groups ($p = \mathbf{0.329}$).
- *Differential abundance:* **28 taxonomic features** (22 named families, one of them archaeal, and 6 unclassified lineages) differed between groups at BH-FDR < 0.25 (**12 enriched** and **16 depleted** in UC), including **Enterobacteriaceae**, **Morganellaceae** and **Neisseriaceae** (enriched) and **Methanobacteriaceae** and **Barnesiellaceae** (depleted).

Take-home message. UC was associated with reduced within-sample diversity and a distinct community composition compared with healthy children, with a differential-abundance profile marked by an expansion of Enterobacteriaceae-related taxa. Given the small cohort, the exploratory significance threshold and the lack of adjustment for treatment, these findings are associations to be validated in larger cohorts.

2 Background and objectives

Ulcerative colitis is a form of inflammatory bowel disease characterized by inflammation and ulceration of the colon. Gut microbial dysbiosis has been reported in both children and adults with UC. This report reproduces and extends, with a fully scripted workflow, the 16S rRNA analysis of the pediatric cohort published by Zuo et al. (2022).

The analysis addresses four questions:

1. Is the sequencing data of sufficient quality, and how many reads and variants are retained after processing?
2. How is the gut microbial community composed at the phylum level?
3. Do within-sample (alpha) and between-sample (beta) diversity differ between UC and healthy children, after accounting for age and gender?
4. Which microbial families differ in abundance between the two groups?

3 Dataset

Table 1: Characteristics of the analyzed cohort.

Feature	Description
Source	NCBI SRA, BioProject PRJNA759642 (Zuo et al., 2022)
Sample type	Fecal samples, pediatric cohort
Samples analyzed	42 (main cohort selected from the 114 runs of the BioProject)
Groups	Healthy (n = 23), UC (n = 19)
Age	Healthy: median 14 (range 8–21) years; UC: median 15 (range 7–20) years
Gender	Healthy: 13 female / 10 male; UC: 9 female / 10 male
Sequencing	Illumina MiSeq, paired-end 2 × 150 bp, 16S rRNA V4 region (515F/806R)

4 Methods

4.1 Raw-read quality control

Raw reads were assessed with FastQC and summarized with MultiQC. Aggregate quality profiles were additionally inspected in DADA2 to set the truncation lengths. Primers had already been removed by the data submitters before SRA deposition, and no adapter or primer contamination was detected.

4.2 ASV inference and taxonomy

Reads were processed with DADA2: filtering and trimming (truncation at 140 bp for forward and 130 bp for reverse reads, maximum 2 expected errors per read, no ambiguous bases, PhiX removal), separate error-rate learning for forward and reverse reads, independent per-sample denoising, paired-end merging, and removal of chimeras with the consensus method. Taxonomy was assigned with the DADA2 naive Bayesian classifier trained on SILVA 138.2 (non-redundant 99% set), followed by species-level assignment by exact matching.

4.3 Community analysis

ASVs, taxonomy and metadata were integrated in a phyloseq object. Analyses were performed at three taxonomic levels: phylum (composition), genus (alpha and beta diversity) and family (differential abundance). ASVs were aggregated with `tax_glom` on the unfiltered table. ASVs without an annotation at the chosen rank were retained as separate “unclassified” units, one for each distinct higher-rank lineage, and are labelled with their lowest assigned rank.

Alpha diversity. The Shannon index was calculated on genus-level counts and compared between groups with the Wilcoxon rank-sum test. The effect of disease status was then estimated with a linear model on log-transformed Shannon diversity, adjusted for age and gender. Sequencing-depth effects were evaluated with Spearman correlations (library size vs Shannon and vs observed genera), by adding log10 library size to the model, and with a rarefaction sensitivity analysis at 1,000, 5,000, 10,000, 30,000, 50,000 and 100,000 reads. At each depth, the Shannon index was computed on each of 100 rarefied tables and averaged per sample; samples with fewer reads than the target depth were retained with all their reads.

Beta diversity. Bray-Curtis dissimilarities were computed on genus-level relative abundances and visualized with PCoA. Group differences were tested with PERMANOVA (`vegan::adonis2`, marginal effects of disease status, age and gender, 999 permutations) and homogeneity of dispersion with `betadisper` and `permutest` (999 permutations). Multiple Regression on distance Matrices (`ecodist::MRM`, 999 permutations) was used to model pairwise dissimilarities as a function of the disease-status combination of each sample pair

(Healthy–UC and UC–UC, with Healthy–Healthy as reference), the gender combination (Female–Male and Female–Female, with Male–Male as reference) and the age difference.

Differential abundance. Family-level counts were filtered (families with zero counts in at least 90% of samples, or with variance below half of the median variance, were removed), normalized with TMM, and tested for UC vs Healthy with the edgeR exact test after estimating negative-binomial dispersions. P-values were adjusted with the Benjamini–Hochberg procedure and taxa with $\text{FDR} < 0.25$ were considered significant. This permissive threshold was adopted because of the small sample size and the exploratory aim of the analysis; the taxa that also passed $\text{FDR} < 0.05$ are reported separately.

All permutation procedures used a fixed seed (1234). The analysis was run in R (v4.6.0); package versions are recorded in `renv.lock` and in the session-information files saved by each script.

5 Results

5.1 Sequencing quality

Raw read counts per sample ranged from **22,274 to 1,592,901 (median 107,241)**. Base quality was generally high along the reads, with localized fluctuations and a drop in the last cycles, and no adapter contamination was observed (Figure 1, Figure 2). In the reverse reads, the proportion of reads reaching each cycle (red line) begins to decline at about cycle 140. Modules flagged as warnings or failures by FastQC (per-base sequence content, duplication levels, overrepresented sequences, GC content) are expected for amplicon data, in which all reads derive from the same short, low-diversity locus, and do not indicate technical problems.

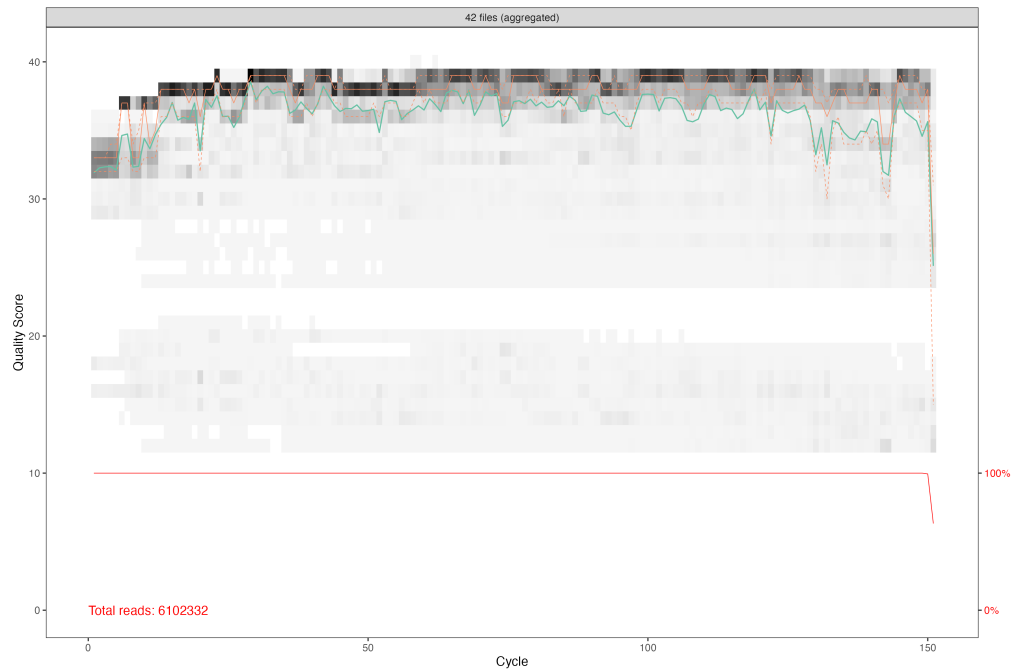


Figure 1: Aggregate read-quality profile of the forward reads (42 files).

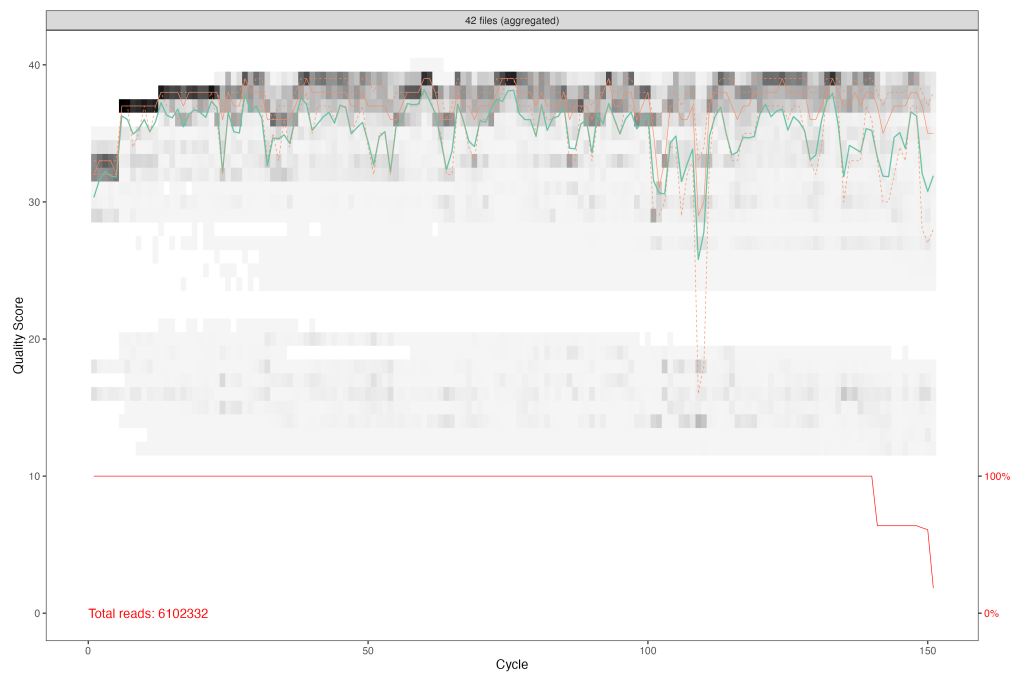


Figure 2: Aggregate read-quality profile of the reverse reads (42 files).

5.2 ASV inference

Truncation lengths (140 bp forward, 130 bp reverse) were set on the aggregate quality profiles to exclude the final cycles, where quality drops and, in the reverse reads, part of the reads end, while leaving enough overlap to merge the V4 amplicon. The learned error models followed the observed error rates (Figure 3, Figure 4), and about 90% of denoised reads were successfully merged (89.7%, ratio of medians; Table 2).

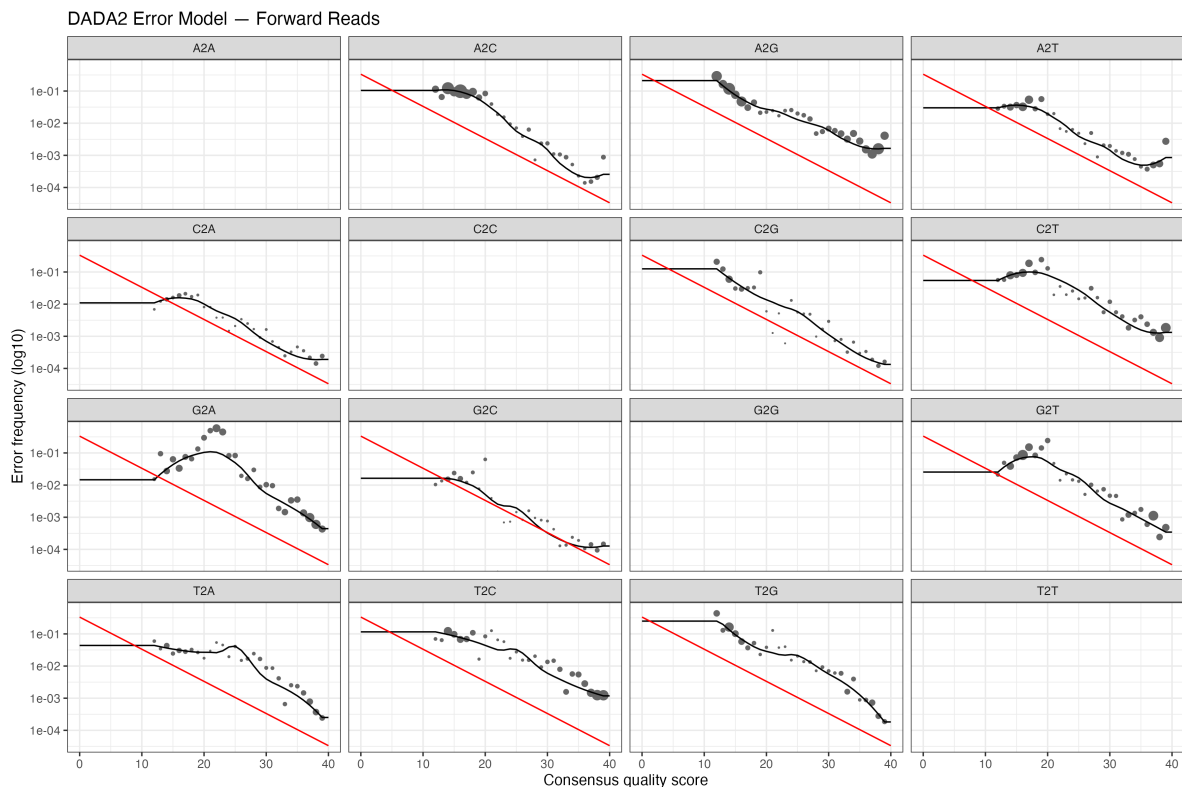


Figure 3: DADA2 error model for the forward reads. Points are observed error rates, black lines the fitted model, red lines the nominal Q-score expectation.

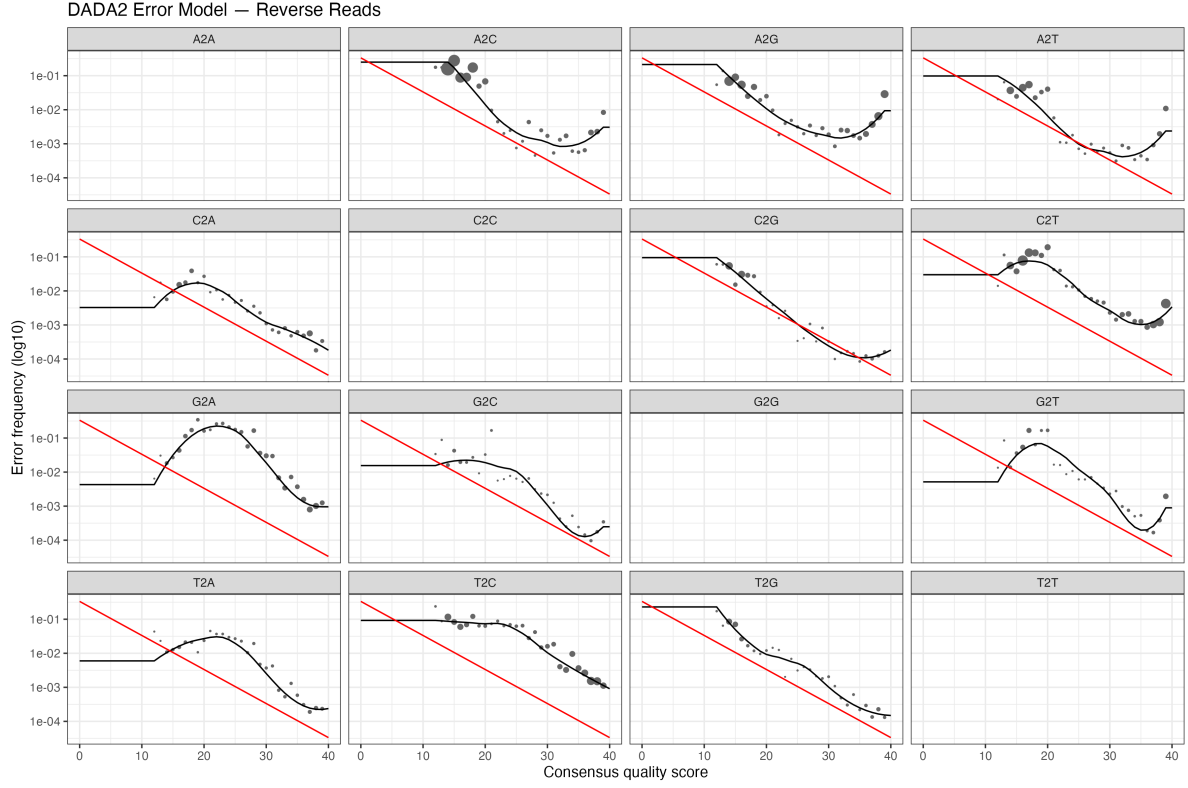


Figure 4: DADA2 error model for the reverse reads.

Table 2 summarizes how reads were retained along the pipeline. In total, 3,564 non-chimeric ASVs were obtained across 42 samples; 20.57% of merged reads were identified as chimeric and removed, and overall 73.42% of all input reads (summed across samples) were retained.

Table 2: Read tracking through the DADA2 pipeline. The last column is the median of each step divided by the median input, and therefore differs from the overall retention computed on summed reads (73.42%).

Step	Median reads per sample (range)	Median reads, % of median input
Input	107,241 (22,274–1,592,901)	100
Filtered	106,790 (22,274–1,577,430)	99.58
Denoised (forward / reverse)	105,622 / 105,414 (22,009–1,570,370 / 22,061–1,568,585)	98.49 / 98.30
Merged	94,729 (20,363–1,503,715)	88.33
Non-chimeric	73,465 (13,639–1,216,057)	68.50

5.3 Taxonomic assignment and composition

Of the 3,564 ASVs, 99.4% were classified at phylum level, 90.6% at family level, 67.8% at genus level and 7.2% at species level.

Sequencing depth after processing ranged from 13,639 to 1,216,057 reads per sample (mean 106,668; Figure 5). ASV prevalence was low for most variants (median prevalence: 1 sample out of 42; Figure 6), as is typical of sparse microbiome tables.

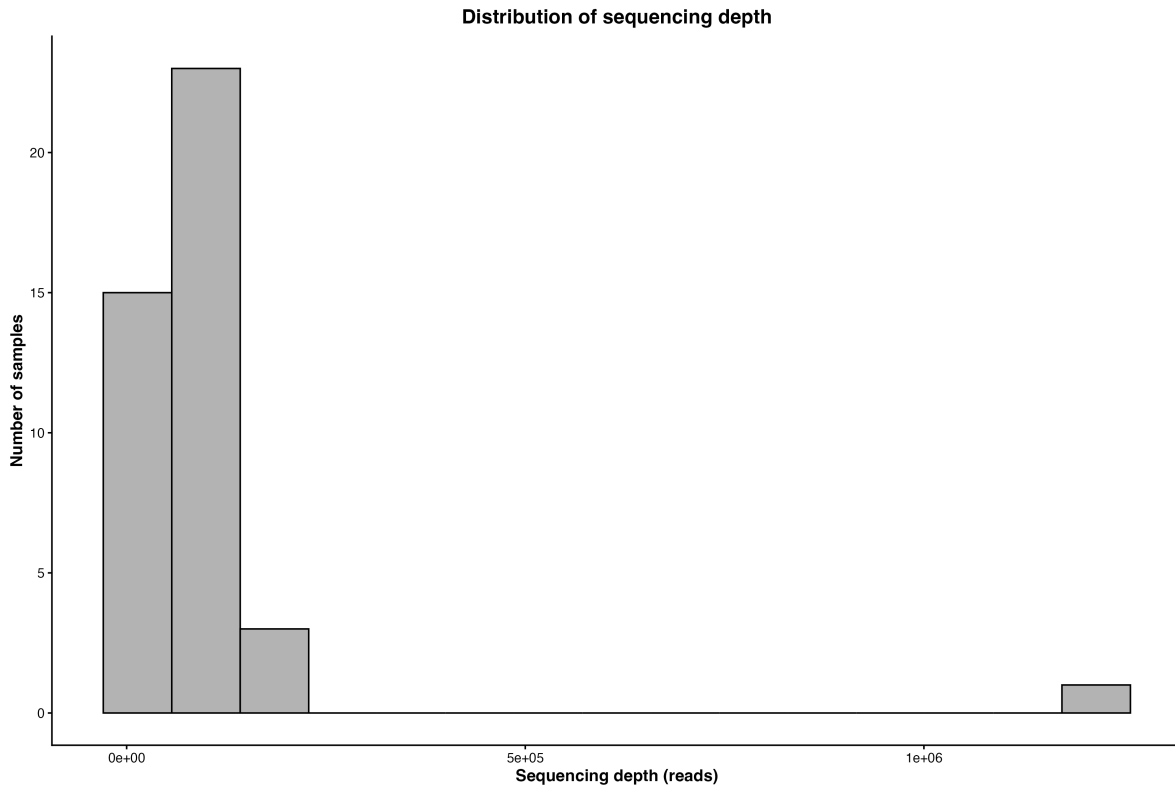


Figure 5: Distribution of sequencing depth (reads per sample after processing).

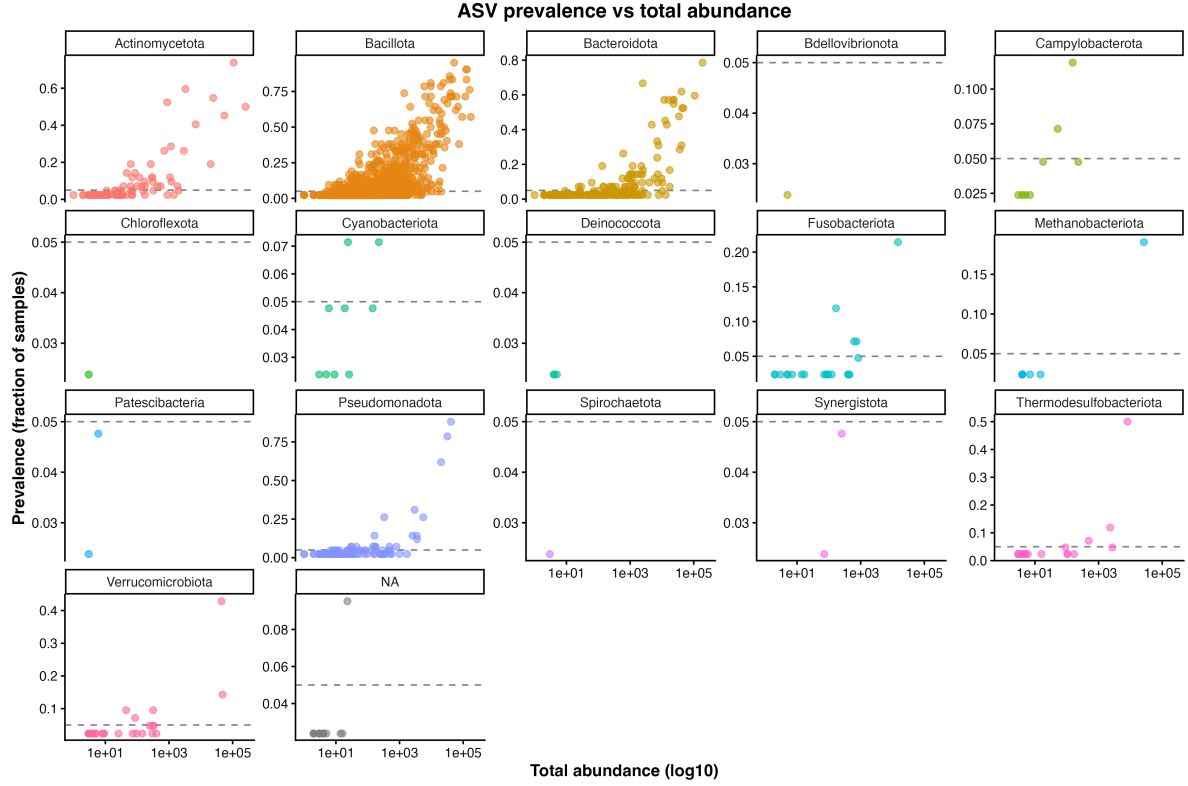


Figure 6: ASV prevalence versus total abundance, by phylum. The dashed line marks 5% of samples.

At phylum level the community was dominated by four phyla with mean relative abundance 2% (Figure 7, Table 3). Compared with Healthy samples, UC samples showed a lower proportion of Bacillota (52.2% vs 65.5%) and Actinomycetota (5.06% vs 9.21%), together with a higher proportion of Bacteroidota (30.2% vs 20.2%) and Pseudomonadota (9.49% vs 1.21%). These are descriptive means; group differences were tested at family level (see Differential abundance).

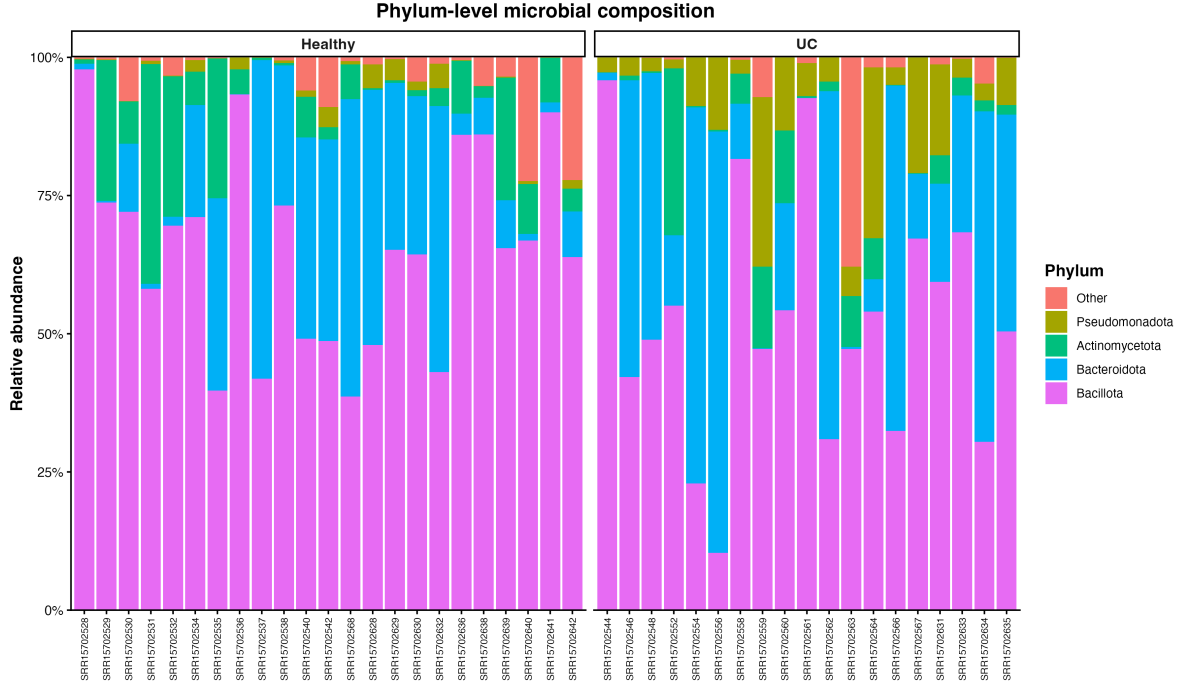


Figure 7: Phylum-level composition of each sample, grouped by clinical status. Phyla below 2% mean relative abundance are grouped as ‘Other’.

Table 3: Mean phylum-level relative abundance by group.

Phylum	Healthy, mean relative abundance (%)	UC, mean relative abundance (%)
Bacillota	65.5	52.2
Bacteroidota	20.2	30.2
Actinomycetota	9.21	5.06
Pseudomonadota	1.21	9.49

5.4 Alpha diversity

Shannon diversity was lower in UC samples (median 2.442, IQR 2.031–2.686) than in healthy samples (median 2.693, IQR 2.564–3.044); Wilcoxon $p = 0.00205$ (Figure 8).

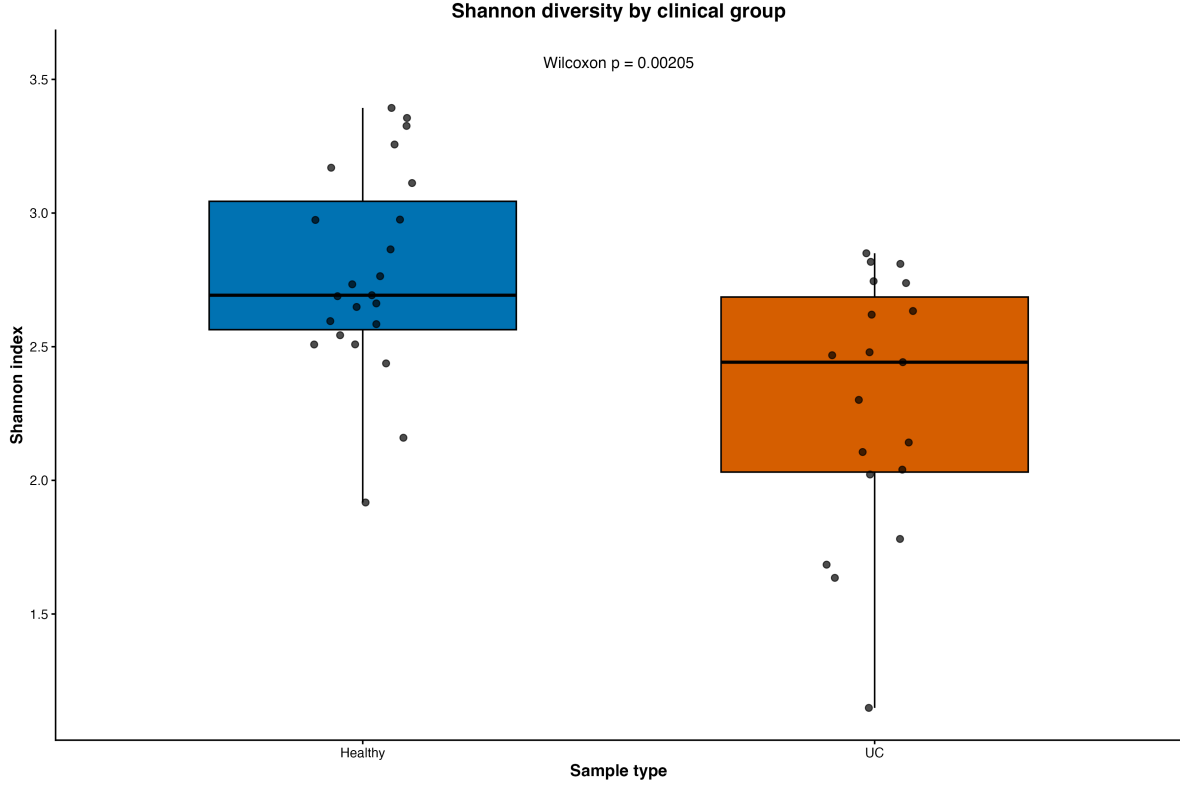


Figure 8: Genus-level Shannon diversity by clinical group (Wilcoxon rank-sum test).

In the linear model adjusted for age and gender (Table 4), UC was associated with a 0.804-fold change in Shannon diversity relative to healthy samples (95% CI 0.715–0.904; $p = 0.000570$). Age was not associated with diversity ($p = 0.653$). Male gender showed a borderline, non-significant association with higher diversity (fold change 1.12; $p = 0.0519$).

Table 4: Linear model of $\log(\text{Shannon})$ on disease status, age and gender ($R^2 = 0.314$).

Term	Estimate (log scale)	95% CI	Fold change	p-value
UC vs Healthy	-0.2182	-0.336 to -0.101	0.804	0.000570
Age (per year)	-0.0038	-0.021 to 0.013	0.996	0.653
Gender (male vs female)	0.1159	-0.001 to 0.233	1.123	0.0519

5.4.1 Robustness to sequencing depth

Library size was moderately correlated with Shannon diversity (Spearman $\rho = 0.425$, $p = 0.00499$) and strongly correlated with the number of observed genera ($\rho = 0.734$, $p < 0.001$). When $\log_{10}$ library size was added to the model, the UC effect was attenuated but remained statistically significant (estimate = -0.181, $p = 0.00984$). Rarefaction curves (Figure 9) reached a plateau for most samples.

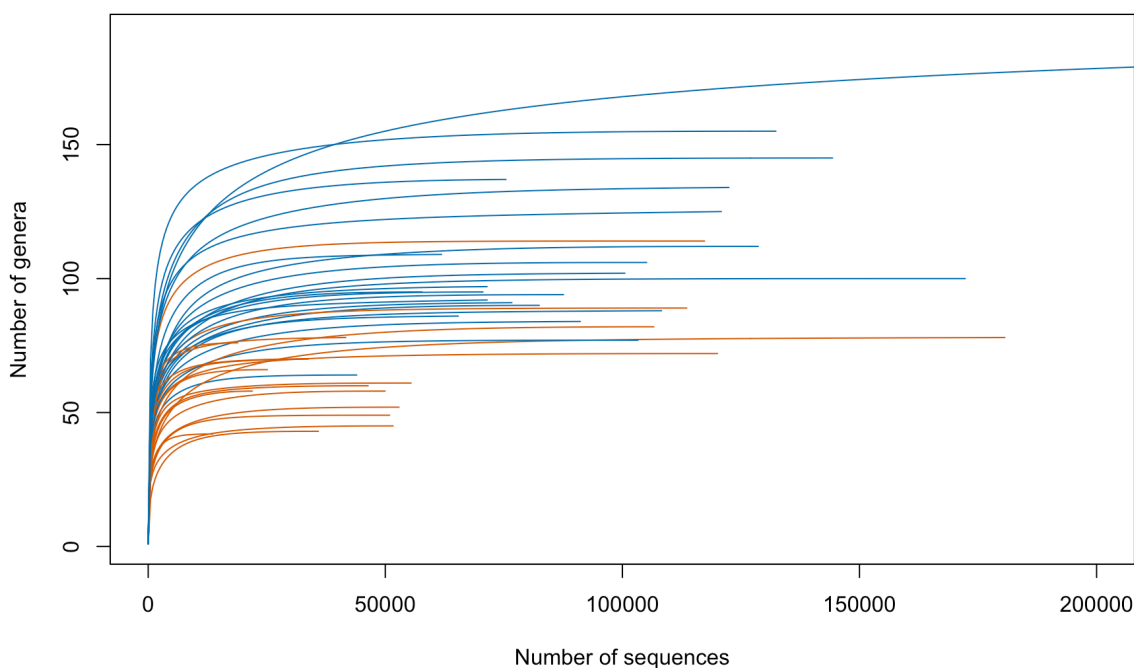


Figure 9: Genus-level rarefaction curves (blue: Healthy, orange: UC).

The group comparison was repeated after rarefaction at six depths (Figure 10, Table 5). Mean Shannon diversity was slightly lower at 1,000 reads and stable from 5,000 reads upward, and the difference between Healthy and UC was preserved at every depth (Wilcoxon $p = 0.00205$ at all depths; the test depends only on the ranking of samples, which did not change). All samples have at least 13,639 reads, so the comparisons at 1,000, 5,000 and 10,000 reads are fully rarefied; at 30,000, 50,000 and 100,000 reads, 4, 11 and 26 samples fall below the target depth and are kept with all their reads, so those panels are only partly rarefied.

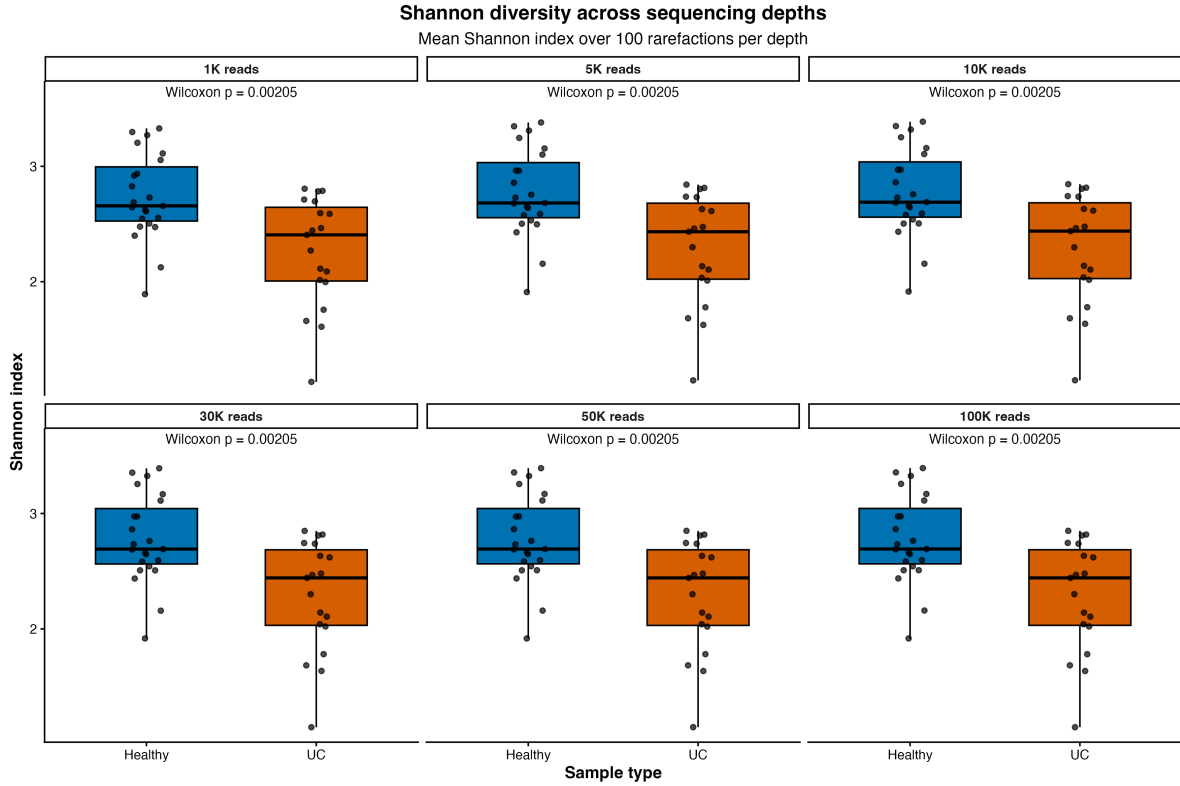


Figure 10: Shannon diversity by clinical group at increasing rarefaction depths. For each sample, the index was calculated on each of 100 rarefied tables and averaged.

Table 5: Group comparison of Shannon diversity across rarefaction depths.

Depth (reads)	Samples below depth (retained in full)	Wilcoxon p-value
1,000	0	0.00205
5,000	0	0.00205
10,000	0	0.00205
30,000	4	0.00205
50,000	11	0.00205
100,000	26	0.00205

5.5 Beta diversity

The first two PCoA axes of the genus-level Bray-Curtis distances explained 30.3% and 13.1% of the variation (Figure 11). Samples partly separated by clinical group.

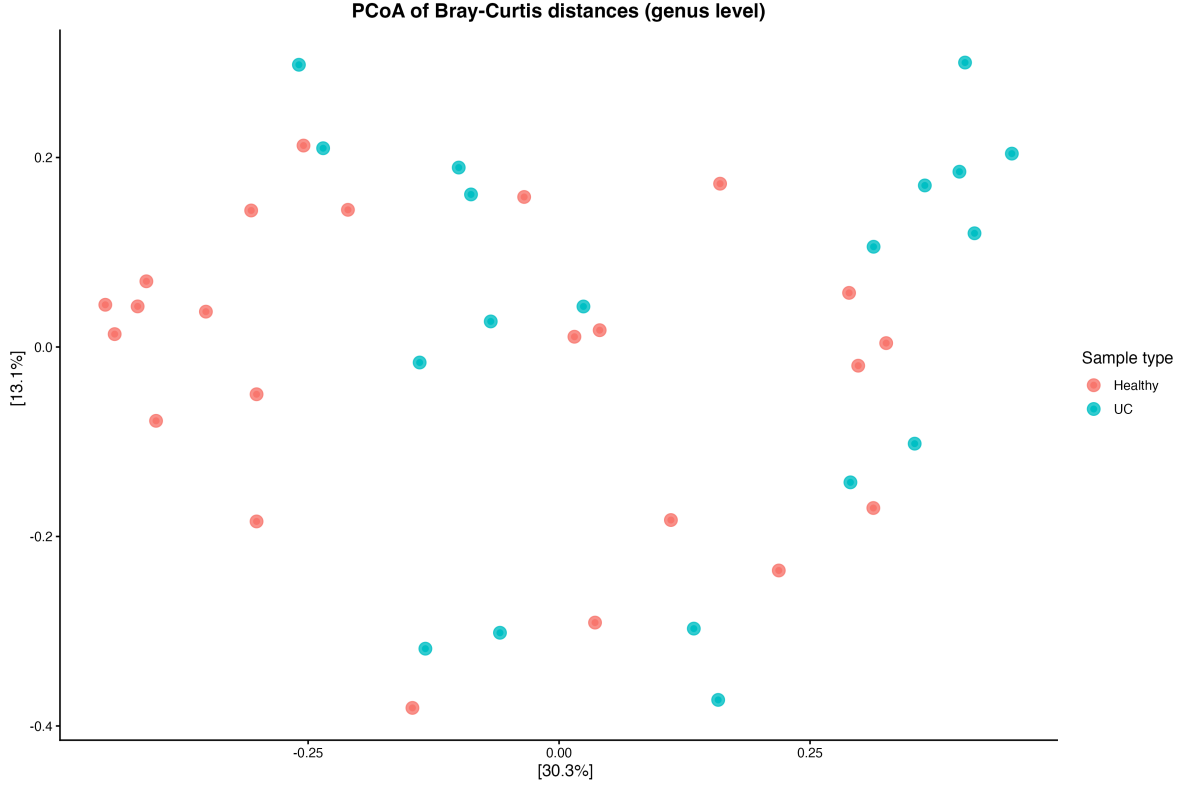


Figure 11: PCoA of genus-level Bray-Curtis dissimilarities, coloured by clinical group.

PERMANOVA with marginal effects (Table 6) showed that disease status explained 9.70% of the variation (pseudo- $F = 4.44$, $p = 0.001$) after adjustment for age and gender. Age was not significantly associated with community composition ($R^2 = 4.11\%$, $p = 0.071$), and neither was gender ($R^2 = 3.57\%$, $p = 0.104$).

Table 6: PERMANOVA on genus-level Bray-Curtis dissimilarities, marginal effects, 999 permutations.

Term	df	Sum of squares	R^2	pseudo-F	p-value
Disease status	1	1.0468	0.0970	4.4435	0.001
Age	1	0.4429	0.0411	1.8802	0.071
Gender	1	0.3849	0.0357	1.6336	0.104
Residual	38	8.9522	0.8297	—	—

Because PERMANOVA is sensitive to differences in within-group variability, homogeneity of multivariate dispersion was tested: dispersion did not differ significantly between groups ($F =$

0.936, $p = 0.329$; Figure 12), although UC samples show a visibly wider spread and the power of this test is limited with 19 UC samples.

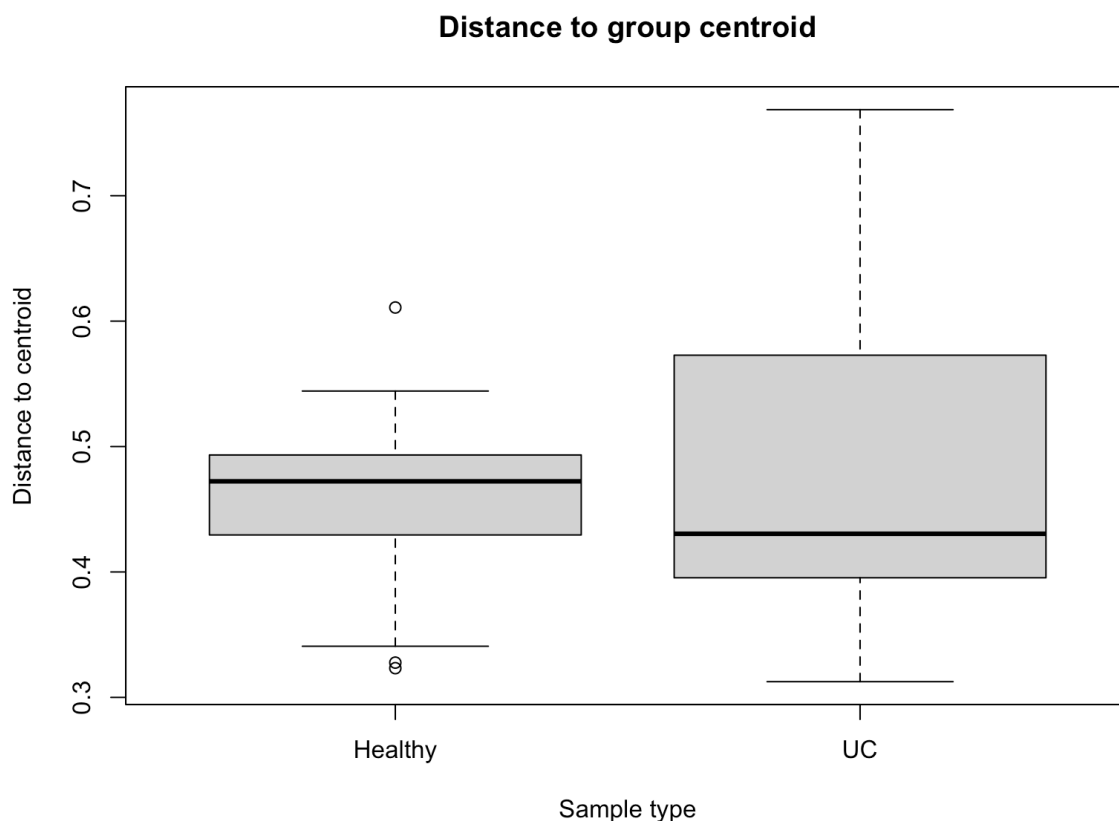


Figure 12: Distance to the group centroid (Bray-Curtis) by clinical group.

Multiple Regression on distance Matrices (Table 7) gives a complementary view, modelling pairwise dissimilarities. Healthy–UC pairs were more dissimilar than Healthy–Healthy pairs (coefficient 0.0817, $p = 0.007$). UC–UC pairs had a positive coefficient of similar magnitude (0.0600) that did not reach significance ($p = 0.222$); with 19 UC samples, the data therefore do not allow us to conclude whether UC microbiomes are more heterogeneous among themselves than healthy ones. The overall MRM model explained 4.88% of the variation in pairwise dissimilarities and was not significant ($p = 0.122$), so the Healthy–UC coefficient should be read together with the PERMANOVA result.

Table 7: MRM on genus-level Bray-Curtis dissimilarities, 999 permutations.

Predictor (pairwise)	Coefficient	p-value
Healthy-UC pair (vs Healthy-Healthy)	0.0817	0.007
UC-UC pair (vs Healthy-Healthy)	0.0600	0.222
Age difference	-0.0046	0.256
Female-Male pair (vs Male-Male)	0.0119	0.649
Female-Female pair (vs Male-Male)	0.0163	0.724

5.6 Differential abundance

After aggregation at family level, 112 family-level units were available; 75 remained after removing units with zero counts in at least 90% of samples and 59 after the variance filter. TMM normalization factors ranged from 0.134 to 3.985; because many families have zero counts in many samples, the log-CPM distributions are flat at the lower end and the visible effect of normalization is modest (Figure 13, Figure 14). The common dispersion estimate was 6.015 (biological coefficient of variation 2.45), reflecting the strong overdispersion typical of sparse family-level counts.

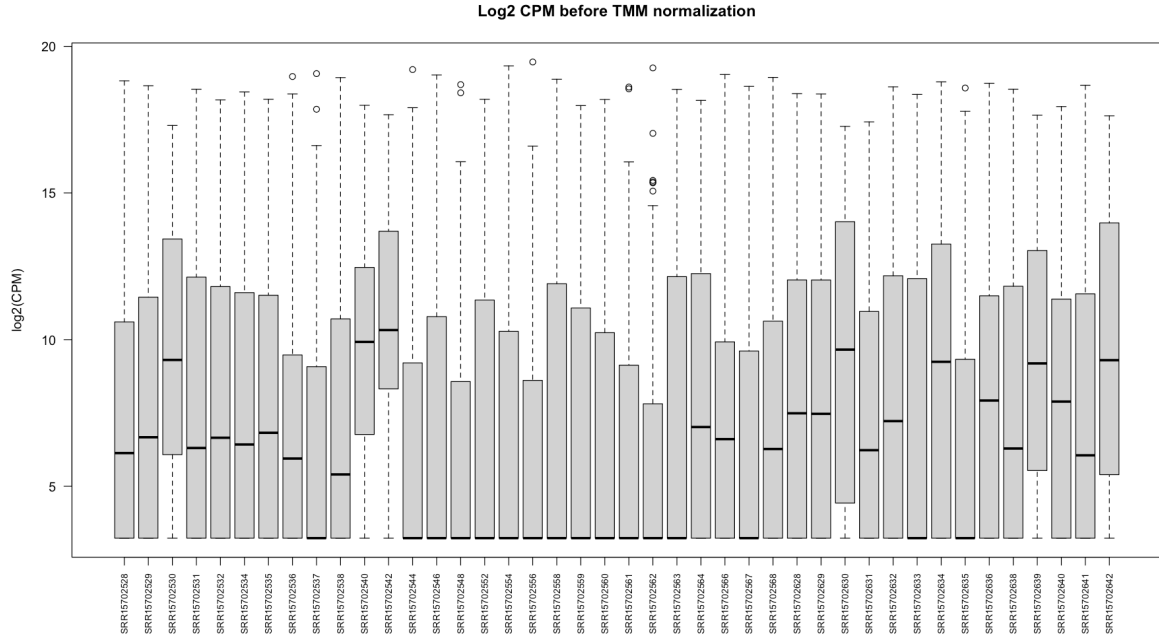


Figure 13: Log2 counts-per-million before TMM normalization.

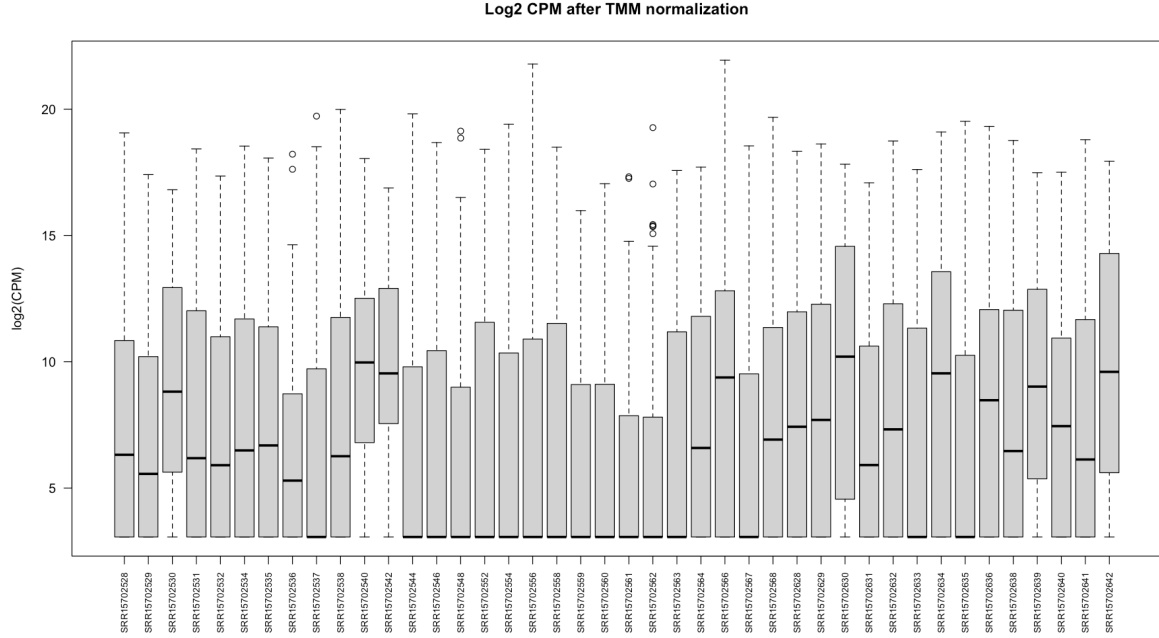


Figure 14: Log2 counts-per-million after TMM normalization.

Exploratory ordinations of the normalized data (MDS and PCA; Appendix) showed a partial separation between groups.

The exact test identified 28 taxa with BH-FDR < 0.25 : 12 enriched and 16 depleted in UC (Figure 15, Table 8, Figure 16). Sixteen of them also passed FDR < 0.05 . The strongest signals included enrichment of Enterobacteriaceae, Morganellaceae, Neisseriaceae, Porphyromonadaceae and Fusobacteriaceae in UC, together with depletion of Methanobacteriaceae, Barnesiellaceae and the [Eubacterium] coprostanoligenes group.

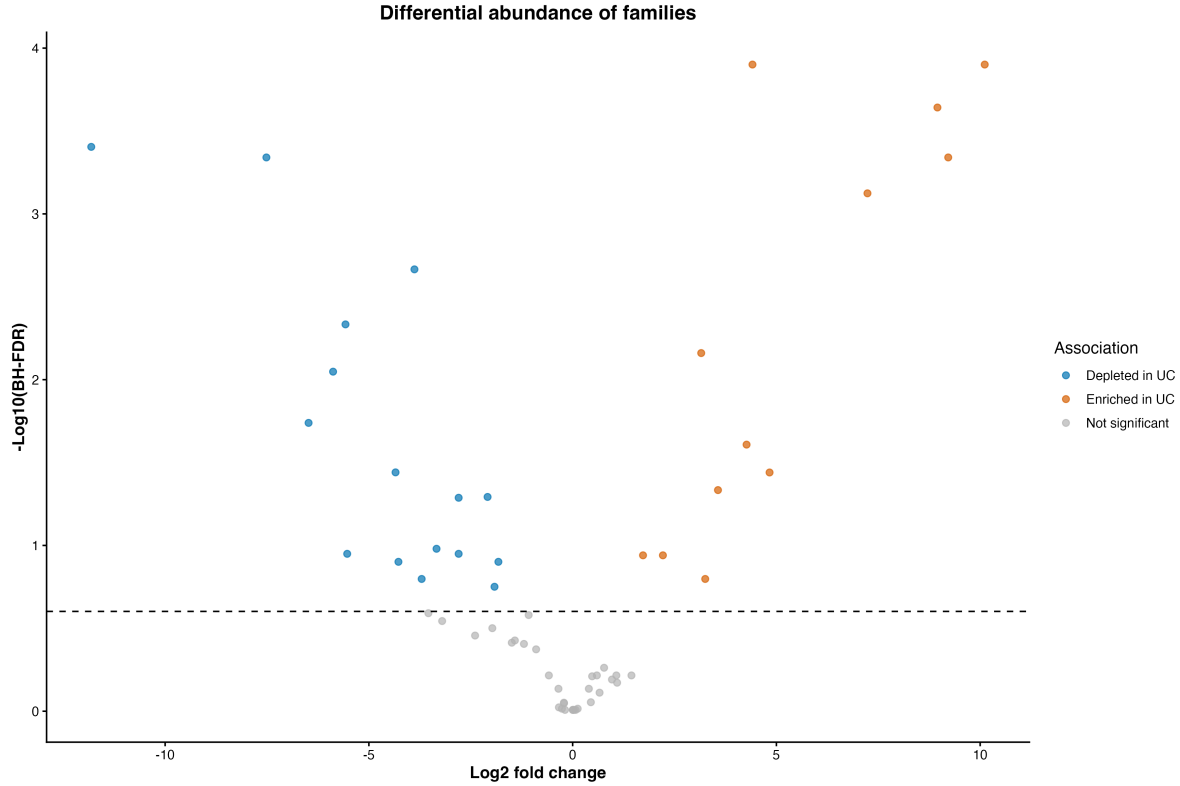


Figure 15: Differential abundance of microbial families, UC vs Healthy. Points above the dashed line have BH-FDR < 0.25.

Table 8: Taxa differentially abundant between UC and Healthy (BH-FDR < 0.25), ordered by FDR. Unclassified lineages are labelled with their lowest assigned rank.

Taxon	log2 FC (UC vs Healthy)	Average log2 CPM	p-value	BH-FDR	Direction
Enterobacteriaceae	4.41	14.76	3.07e-06	0.000126	Enriched in UC
Morganellaceae	10.11	9.05	4.26e-06	0.000126	Enriched in UC
Neisseriaceae	8.95	9.46	1.16e-05	0.000228	Enriched in UC
Methanobacteriaceae	-11.81	11.02	2.67e-05	0.000394	Depleted in UC
Barnesiellaceae	-7.52	12.02	4.28e-05	0.000457	Depleted in UC

Table 8: Taxa differentially abundant between UC and Healthy (BH-FDR < 0.25), ordered by FDR. Unclassified lineages are labelled with their lowest assigned rank.

Taxon	log2 FC (UC vs Healthy)	Average log2 CPM	p-value	BH-FDR	Direction
Porphyromonadaceae	9.22	12.17	4.64e-05	0.000457	Enriched in UC
Fusobacteriaceae	7.23	12.52	8.92e-05	0.000751	Enriched in UC
[Eubacterium] coprostanoligenes group	-3.88	11.98	0.000293	0.00216	Depleted in UC
Unclassified (order Chloroplast)	-5.57	6.48	0.000708	0.00464	Depleted in UC
Pasteurellaceae	3.15	13.07	0.00117	0.00691	Enriched in UC
Moraxellaceae	-5.88	5.67	0.00167	0.00895	Depleted in UC
Victivallaceae	-6.48	6.09	0.00371	0.0182	Depleted in UC
Family XI	4.27	10.18	0.00543	0.0247	Enriched in UC
Unclassified (order Oscillospirales)	-4.35	6.82	0.0086	0.0362	Depleted in UC
Unclassified (order Bacteroidales)	4.83	9.41	0.00923	0.0363	Enriched in UC
Campylobacteraceae	3.57	6.48	0.0126	0.0464	Enriched in UC
Eggerthellaceae	-2.09	11.80	0.0147	0.051	Depleted in UC
Monoglobaceae	-2.80	11.53	0.0157	0.0516	Depleted in UC
Akkermansiaceae	-3.34	14.07	0.0337	0.105	Depleted in UC
Pseudomonadaceae	-2.80	6.62	0.0389	0.112	Depleted in UC
Unclassified (order RF39)	-5.53	12.29	0.04	0.112	Depleted in UC
Bacteroidaceae	1.73	18.56	0.0441	0.115	Enriched in UC
Sutterellaceae	2.21	14.25	0.0447	0.115	Enriched in UC

Table 8: Taxa differentially abundant between UC and Healthy (BH-FDR < 0.25), ordered by FDR. Unclassified lineages are labelled with their lowest assigned rank.

Taxon	log2 FC (UC vs Healthy)	Average log2 CPM	p-value	BH-FDR	Direction
Unclassified (order Clostridia UCG-014)	-4.27	13.19	0.0515	0.126	Depleted in UC
Clostridiaceae	-1.82	13.69	0.0532	0.126	Depleted in UC
UCG-010	-3.70	9.53	0.0719	0.159	Depleted in UC
Carnobacteriaceae	3.25	8.81	0.0729	0.159	Enriched in UC
Unclassified (class Clostridia)	-1.92	7.93	0.0842	0.178	Depleted in UC

Of the 28 taxa, 22 are named families and 6 are unclassified lineages (one enriched, annotated at order Bacteroidales; five depleted, annotated at order Chloroplast, Oscillospirales, RF39, Clostridia UCG-014 and class Clostridia), which cannot be interpreted at family level. Two further points call for caution. The order Chloroplast comprises plant-derived organelle sequences, likely of dietary origin, which were not removed before the analysis; and Methanobacteriaceae is an archaeal family amplified by the same primers, so not all depleted taxa are bacterial. Moreover, several of the strongest signals reflect presence/absence patterns visible in the heatmap (Figure 16): Morganellaceae and Neisseriaceae are detected mainly in a subset of UC samples, and Methanobacteriaceae only in a subset of healthy samples. Log2 fold changes of this size (up to about 10) derive from these patterns and should be interpreted with caution.

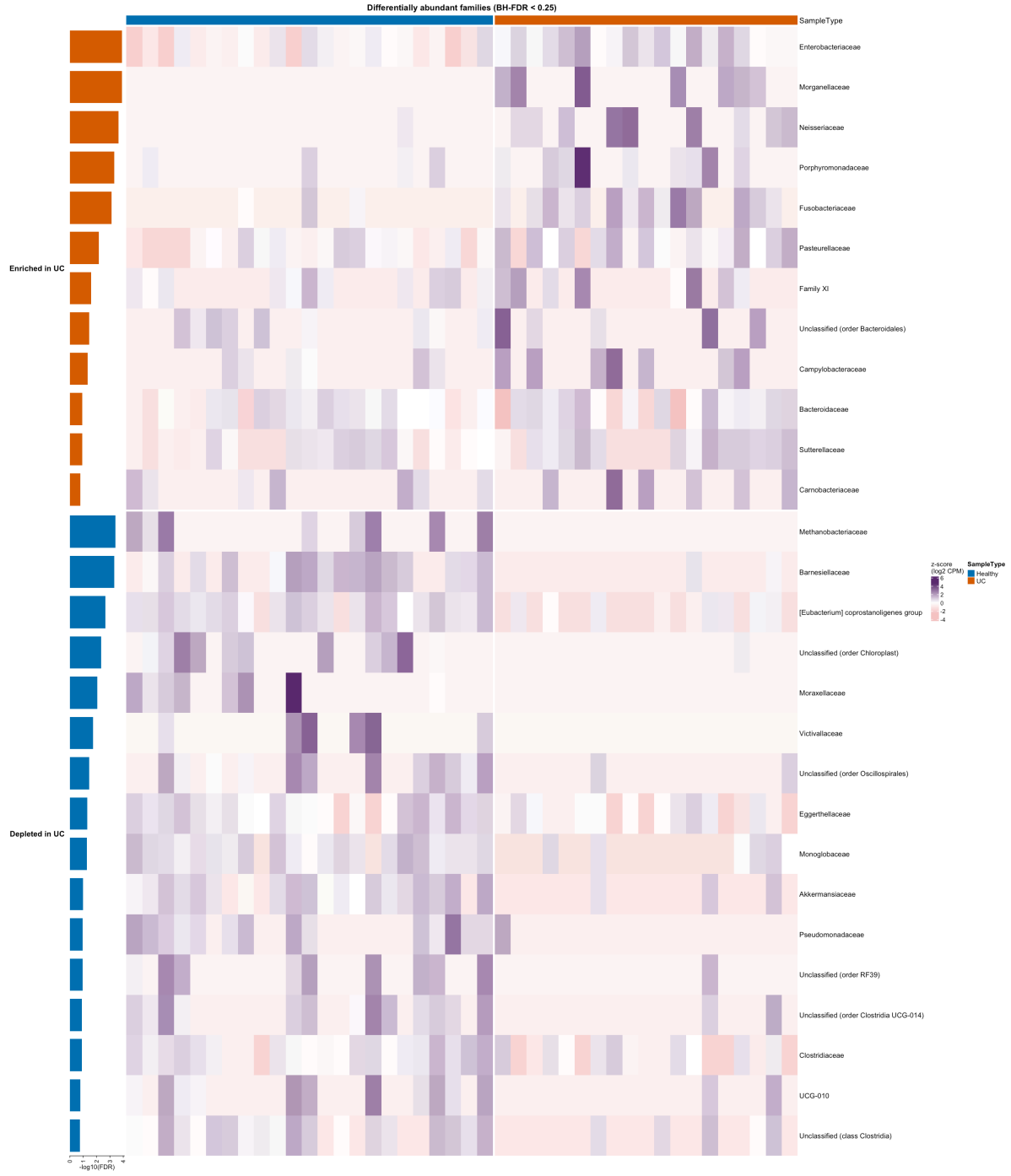


Figure 16: Row-scaled log2-CPM (z-score) of the 28 differentially abundant taxa (BH-FDR < 0.25). Rows are split by direction of change and ordered by FDR within each block; bars show $-\log_{10}(\text{BH-FDR})$. Unclassified lineages are labelled with their lowest assigned rank.

6 Discussion

6.1 Interpretation

The present analysis indicates an alteration of the gut microbiome in children with UC compared with healthy controls. Alpha diversity was significantly lower in UC, with a median Shannon index of 2.44 compared with 2.69 in healthy children (Wilcoxon $p = 0.002$). This difference remained evident after adjustment for age and gender, with UC associated with an approximately 20% lower Shannon diversity (fold change 0.80, 95% CI 0.71–0.90, $p < 0.001$). At the community-composition level, disease status explained 9.7% of the variation in genus-level Bray–Curtis dissimilarities (PERMANOVA pseudo- $F = 4.44$, $p = 0.001$), whereas age and gender did not reach conventional statistical significance. Multivariate dispersion did not differ significantly between groups ($F = 0.94$, $p = 0.329$), which suggests that the PERMANOVA signal mainly reflects differences in community centroids, although the power of the dispersion test is limited.

The taxonomic profiles were consistent with this broader shift in community structure. Healthy samples were dominated by Bacillota and Bacteroidota, whereas UC samples showed a lower relative abundance of Bacillota and higher Bacteroidota and Pseudomonadota. Differential-abundance analysis further identified 28 taxa at BH-FDR < 0.25 (22 named families and 6 unclassified lineages), including 12 enriched and 16 depleted in UC, with 16 passing the more stringent FDR < 0.05 threshold. Among the strongest signals, Enterobacteriaceae, Morganellaceae and Neisseriaceae were enriched in UC, whereas Methanobacteriaceae, Barnesiellaceae and the [Eubacterium] coprostanoligenes group were depleted. The expansion of Enterobacteriaceae-related taxa is consistent with the well-described association between intestinal inflammation and facultative anaerobic Proteobacteria (Pseudomonadota), whereas depletion of taxa belonging to anaerobic and potentially metabolically beneficial groups may reflect disruption of the healthy intestinal microbial ecosystem. These taxonomic changes should nevertheless be interpreted as associations rather than evidence of causality.

Taken together, the alpha-diversity, beta-diversity and differential-abundance results are consistent with UC-associated microbiome dysbiosis. The lower Shannon diversity indicates reduced within-sample community complexity, while PERMANOVA indicates that UC status is also associated with a shift in overall community composition. Differential-abundance analysis identifies specific taxonomic features that accompany this separation, including expansion of Enterobacteriaceae-related taxa and depletion of several anaerobic taxa.

6.2 Comparison with the original study

Zuo et al. (2022) reported that pediatric UC was characterized by a dysbiotic and less diverse gut microbial population, with some species of the Christensenellaceae family depleted and some species of the Enterobacteriaceae family enriched, and reported broadly similar findings

between 16S rRNA and shotgun metagenomic data for alpha diversity, beta diversity and prediction accuracy. In the present analysis, alpha diversity was concordant with these findings, with significantly lower Shannon diversity in UC, and community composition was also concordant, with disease status significantly associated with genus-level Bray–Curtis community structure. Enterobacteriaceae were among the differentially abundant families in the present analysis and were strongly enriched in UC ($\log_2$ fold change = 4.41, BH-FDR = 1.26×10^{-5}). In contrast, Christensenellaceae were not differentially abundant in the present analysis ($P = 0.229$, BH-FDR = 0.386), and therefore the reported Christensenellaceae depletion could not be reproduced at the family level in this analysis.

Differences between the studies are expected because the present analysis tested differential abundance at the family level, whereas Zuo et al. also resolved taxonomic differences at the species level. In addition, the analytical frameworks differed, including TMM normalization and the edgeR exact test used here, as well as differences in feature filtering and potentially in the reference taxonomy. Consequently, absence of a significant Christensenellaceae signal in the present family-level analysis does not necessarily contradict species-level depletion reported by Zuo et al.; species-specific changes can be obscured when taxa are aggregated to a higher taxonomic level.

6.3 Limitations

- **Sample size.** With 42 children, power is limited, in particular for the family-level tests, where an FDR threshold of 0.25 was used. Results should be considered exploratory and in need of validation.
- **Cross-sectional design and confounding.** The data come from a single time point. Age and gender were accounted for in alpha and beta diversity, but not in the differential abundance test, which compares two groups only. Treatment-related variables available in the metadata (antibiotics, biologics, steroids, aminosacylates, immunomodulators) were not modelled and may influence the UC microbiome.
- **Compositionality.** 16S data are relative. Differences in the abundance of one family can reflect changes in other families; TMM normalization mitigates but does not remove this limitation, and results are interpreted as differences in observed relative abundance, not absolute abundance.
- **Sparse signals.** Several differentially abundant taxa are detected in only a subset of samples of one group and absent in the other; the corresponding fold changes are large but statistically fragile.
- **Non-bacterial and non-microbial sequences.** Plant organelle (chloroplast) sequences were not filtered out, and the primers also amplify archaea; one significant taxon belongs to each category.
- **Resolution of the V4 region.** Taxonomic assignment is reliable mainly to genus level; species-level calls are exact matches and should be interpreted with caution.

- **Taxonomic aggregation.** ASVs without annotation at the chosen rank are retained as separate unclassified units (6 of the 28 significant taxa), and the unclassified fraction (32.21% of ASVs at genus level) cannot be interpreted biologically.
- **Distance choice.** Beta diversity was based on Bray-Curtis dissimilarities of relative abundances; phylogeny-based metrics (UniFrac) and compositional distances (Aitchison) were not used.

6.4 Possible extensions

- Removal of chloroplast and mitochondrial sequences, and separate handling of archaea, before the diversity and differential-abundance analyses.
- Differential abundance with covariate adjustment (age, gender, treatment), for example an edgeR generalized linear model or a compositional method, as a sensitivity analysis.
- Genus-level differential abundance for a more direct comparison with the original study.
- Phylogenetic tree construction and UniFrac distances.
- Integration with the shotgun metagenomic data of the same cohort.

7 Conclusions

- **Alpha diversity:** UC children showed significantly lower genus-level Shannon diversity than healthy children, with the difference remaining significant after adjustment for age, gender and sequencing depth, and at every rarefaction depth tested.
- **Community composition:** UC was associated with a distinct genus-level community composition, explaining approximately 9.7% of Bray-Curtis variation after accounting for age and gender; dispersion did not differ significantly between groups.
- **Differential abundance:** family-level analysis identified 28 differentially abundant taxa (22 named families and 6 unclassified lineages) at BH-FDR < 0.25 , 12 enriched and 16 depleted in UC; 16 also met the more stringent FDR < 0.05 threshold. Enterobacteriaceae-related taxa were among the strongest UC-enriched signals, whereas Methanobacteriaceae and Barnesiellaceae were among the strongest depleted taxa.
- **Overall microbiome profile:** taken together, the results support a less diverse and compositionally distinct gut microbiome in pediatric UC. The findings are exploratory and should be validated in larger, independent cohorts with more comprehensive adjustment for treatment and other clinical confounders.

8 Reproducibility

The analysis is fully scripted and organized in six modules (`download_16S_fastq.sh`, `qc.sh`, `01_dada2.r`, `02_phyloseq_beta_diversity.r`, `03_phyloseq_alpha_diversity.r`,

04_differential_abundance_analysis.r). The R environment is managed with `renv`, random seeds are fixed, and the session information of each script is saved in the log directory. Instructions to rerun the analysis are in the repository README.

9 References

- Zuo W, Wang B, Bai X, et al. (2022). 16S rRNA and metagenomic shotgun sequencing data revealed consistent patterns of gut microbiome signature in pediatric ulcerative colitis. *Scientific Reports*. DOI: 10.1038/s41598-022-07995-7.
- Callahan BJ, McMurdie PJ, Rosen MJ, et al. (2016). DADA2: High-resolution sample inference from Illumina amplicon data. *Nature Methods*, 13:581–583.
- Quast C, Pruesse E, Yilmaz P, et al. (2013). The SILVA ribosomal RNA gene database project. *Nucleic Acids Research*, 41:D590–D596.
- McMurdie PJ, Holmes S. (2013). phyloseq: An R package for reproducible interactive analysis and graphics of microbiome census data. *PLoS ONE*, 8:e61217.
- McMurdie PJ, Holmes S. (2014). Waste not, want not: why rarefying microbiome data is inadmissible. *PLoS Computational Biology*, 10:e1003531.
- Anderson MJ. (2001). A new method for non-parametric multivariate analysis of variance. *Austral Ecology*, 26:32–46.
- Lichstein JW. (2007). Multiple regression on distance matrices: a multivariate spatial analysis tool. *Plant Ecology*, 188:117–131.
- Robinson MD, McCarthy DJ, Smyth GK. (2010). edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics*, 26:139–140.
- Robinson MD, Oshlack A. (2010). A scaling normalization method for differential expression analysis of RNA-seq data. *Genome Biology*, 11:R25.
- Gloor GB, Macklaim JM, Pawlowsky-Glahn V, Egozcue JJ. (2017). Microbiome datasets are compositional: and this is not optional. *Frontiers in Microbiology*, 8:2224.

10 Appendix

10.1 Supplementary figures

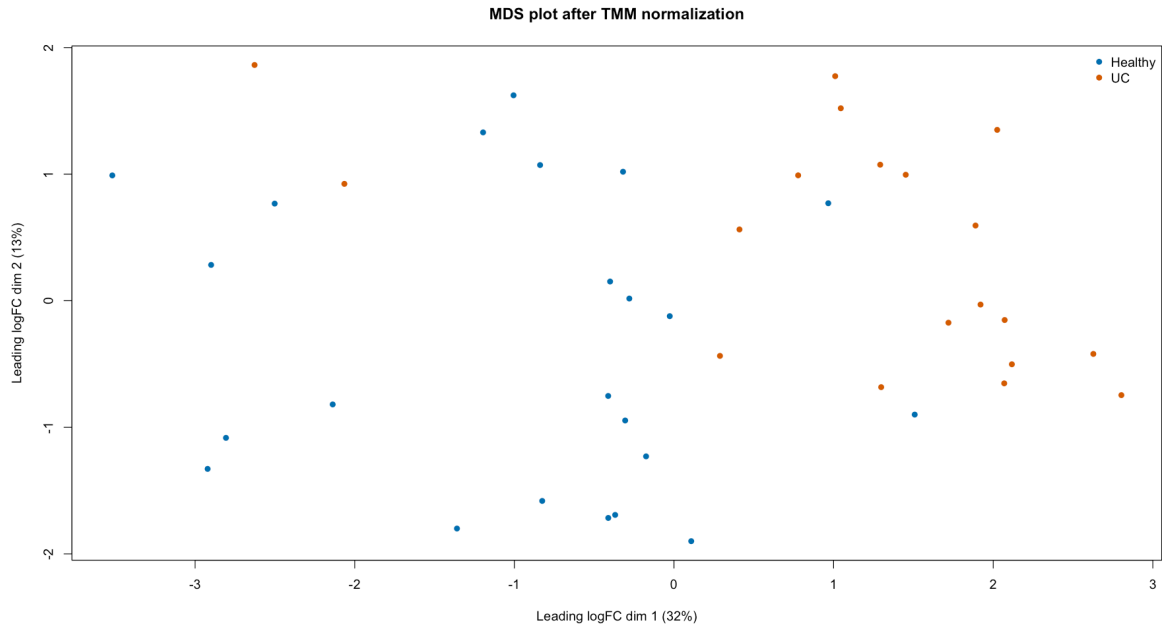


Figure 17: MDS of the TMM-normalized family-level data.

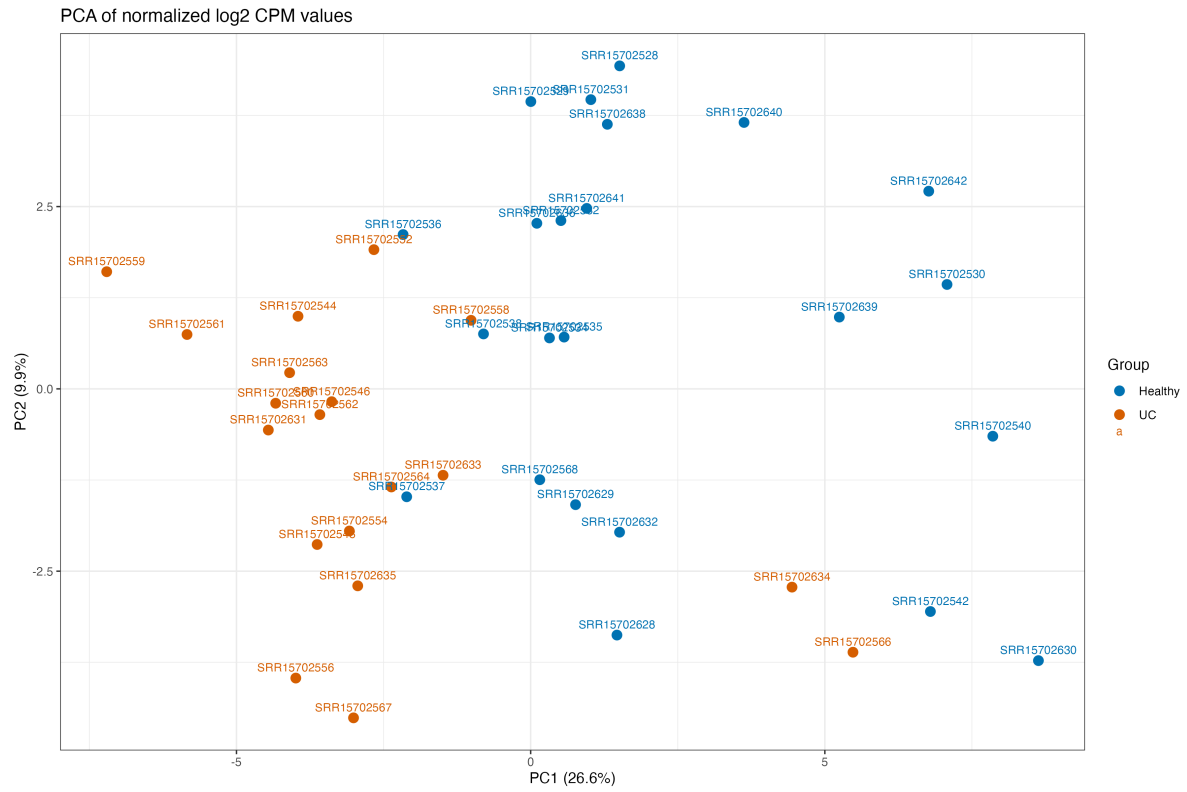


Figure 18: PCA of the normalized log2-CPM values.

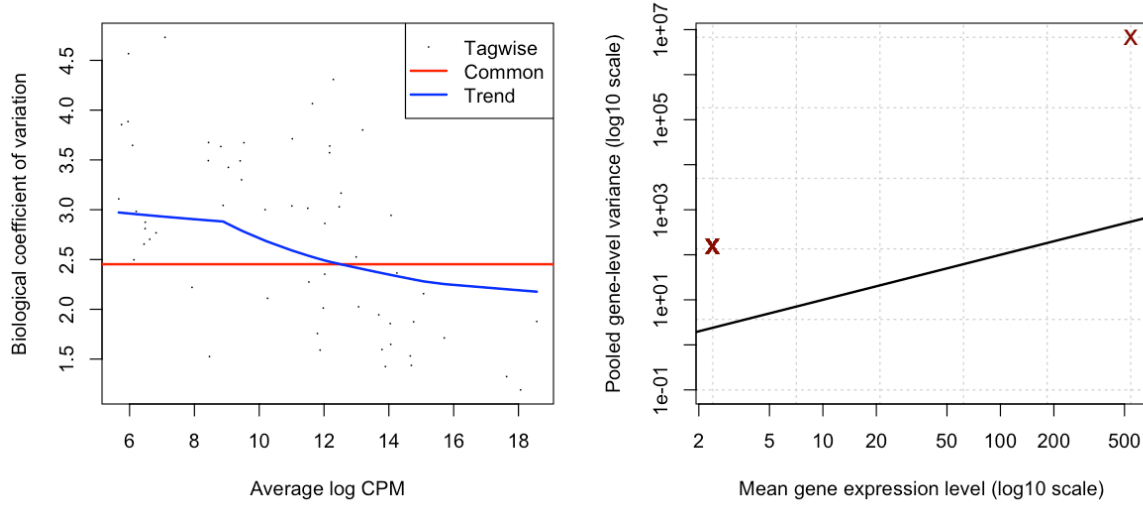


Figure 19: Biological coefficient of variation (left) and mean-variance relationship (right) from the edgeR dispersion estimates.

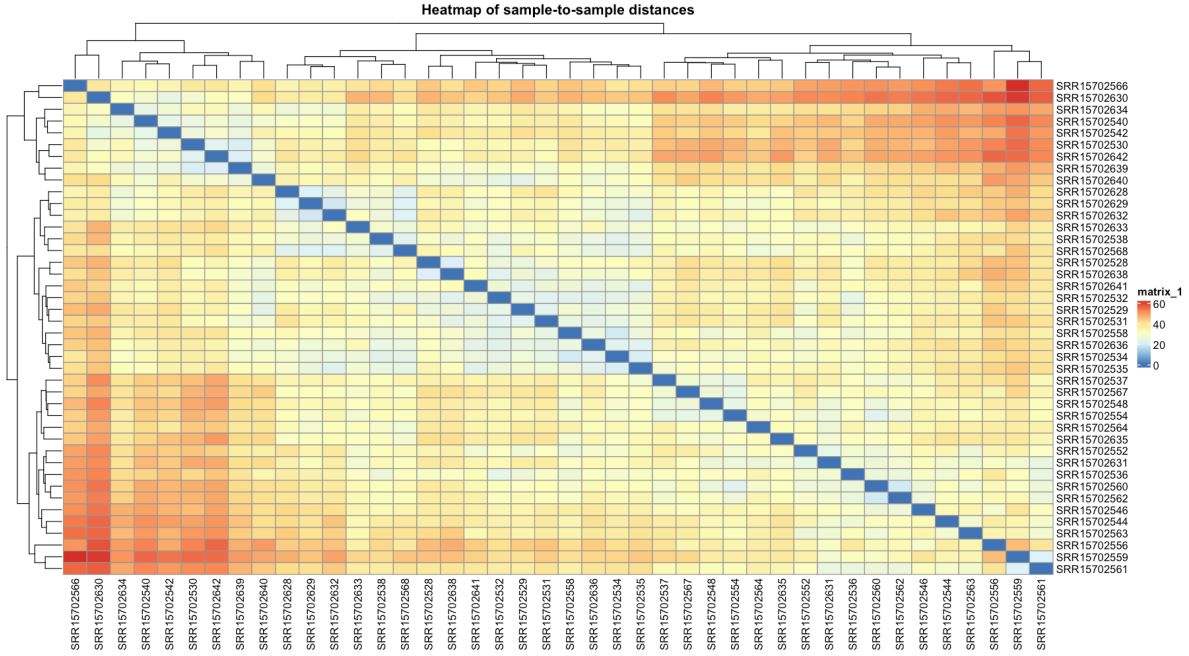


Figure 20: Sample-to-sample Euclidean distances on normalized log2-CPM.

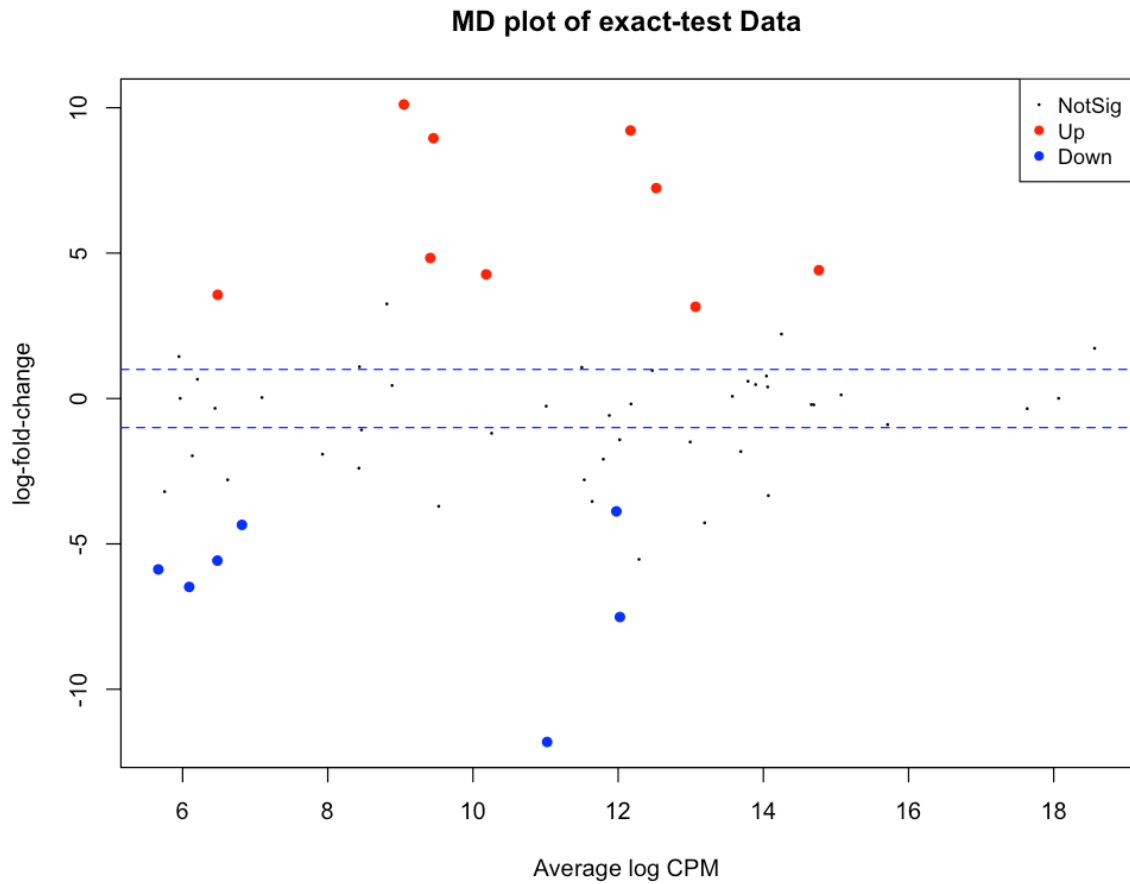


Figure 21: Mean-difference (MD) plot of the exact test.

10.2 Main analysis parameters

Table 9: Main parameters used in the analysis.

Step	Parameter	Value
Filtering	Truncation lengths (F/R)	140 / 130 bp
Filtering	Maximum expected errors (F/R)	2 / 2
Filtering	Ambiguous bases	0
Chimera removal	Method	consensus
Taxonomy	Reference	SILVA 138.2 (nr99 training set + species file)
Alpha diversity	Index / level	Shannon / genus

Step	Parameter	Value
Rarefaction	Depths, repetitions	1K, 5K, 10K, 30K, 50K, 100K; 100
Beta diversity	Distance / level	Bray-Curtis / genus
PERMANOVA, betadisper, MRM	Permutations, seed	999, 1234
Differential abundance	Level / normalization / test	family / TMM / exact test
Differential abundance	Filters	zero counts < 90% of samples; variance $0.5 \times$ median
Differential abundance	Significance	BH-FDR < 0.25

10.3 Software environment

R v4.6.0. The exact versions of all R packages are recorded in `renv.lock`, and the session information of each analysis script is saved in the log directory of the repository.